

Channel-optimized Local Region Riemannian Approach for Motor Imagery Classification

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Abstract—In this paper, we propose a novel motor imagery (MI) classification method that uses channel optimization within local region and Riemannian approach with filter bank to improve classification performance. The proposed method establishes *local regions* as channel subsets consisting of adjacent channels and selects the optimal channels within each region to generate discriminative local Riemannian features. Unlike traditional global channel selection methods that ignore channel location and discard channels completely, the proposed method preserves local information and allows selective inclusion or exclusion of channels in different local regions. Channels are optimized to minimize *confusion area score*, a criterion that assesses the misclassification risk of a Riemannian feature based on Riemannian distance distributions, using a backward iterative algorithm. The EEG signals from each channel-optimized local region are decomposed using a filter bank, and a local covariance matrix is computed for each subband. Several matrices are selected based on the confusion area score, mapped to the tangent space, and fed to a classifier. The performance of the proposed method has been evaluated using the publicly available BCI Competition III dataset IVa, and the results show that the optimization of local region channels improves the performance and achieves higher classification accuracy compared to the related existing methods.

Index Terms—Brain-Computer Interface (BCI), Electroencephalogram (EEG), Motor Imagery (MI), Local region, Channel selection, Riemannian distance, Filter bank

I. INTRODUCTION

Brain-computer interface (BCI) provides a direct communication between human brain and external instruments by decoding brain signals [1]. Electroencephalography (EEG), which non-invasively measures brain signals using electrodes attached on scalp, is mostly utilized in BCI systems due to its convenience and high temporal resolution [2]. Among EEG-based BCI paradigms, motor imagery (MI) has been widely studied and utilized since subjects can voluntarily generate task-related EEG signals without any external stimuli [3].

Common spatial pattern (CSP) [4], which generates spatial filters that maximizes the variance ratio of two classes, is a traditional feature extraction method for EEG-based MI classification. Filter bank CSP (FBCSP), proposed in [5], applies the CSP algorithm to each subband to exploit the

frequency diversity of EEG and achieve performance improvement. However, the CSP algorithm requires estimation of the spatial covariance matrix, and the global CSP approach using whole channels is prone to overfitting problem.

To overcome the overfitting problem, an approach that computes covariance matrices using channel subsets, called local regions [6], consisting of adjacent channels, rather than a global channel set has been investigated. For example, there have been local region CSP [6], which computes local CSPs, and frequency-optimized LRCSP (LRFCS) [7], which combines LRCSP and a filter bank for feature diversity to further improve overall performance.

Recently, a Riemann approach has been studied for MI classification by considering the covariance matrix as a feature, based on the space of the covariance matrix, i.e., the Riemann manifold, and has achieved higher performance than the CSP approach [8], [9]. For example, a method in which a covariance matrix is projected from the Riemannian manifold into tangent space, which is Euclidean space, and fed into linear discriminant analysis (TSLDA) is proposed in [8]. Filter banks and Riemannian tangent space (FBRTS) proposed in [9] computes covariance matrices with multiple time windows and frequency bands to exploit feature diversity.

Inspired by CSP approach based on local region, in [10], a method that combines local region and Riemannian approach has been proposed to improve the performance. The method generates local covariance matrices using local regions and selects several local regions by the criterion designed to evaluate the discriminability of Riemannian feature based on the distribution of Riemannian distance, called confusion area score.

The local region approach improves the performance of the CSP and Riemannian methods. However, some channels belonging to adjacent different local regions may contribute to the specific local region but may not be task-related for others, leading to non-optimal results.

In this paper, we select optimal channels in each local region to improve the MI classification performance. From the EEG signal, spatial covariance matrices are calculated for each local region. For each local region, the proposed method selects several optimal channels to minimize the confusion area score by using backward iterative algorithm. Using the channel-

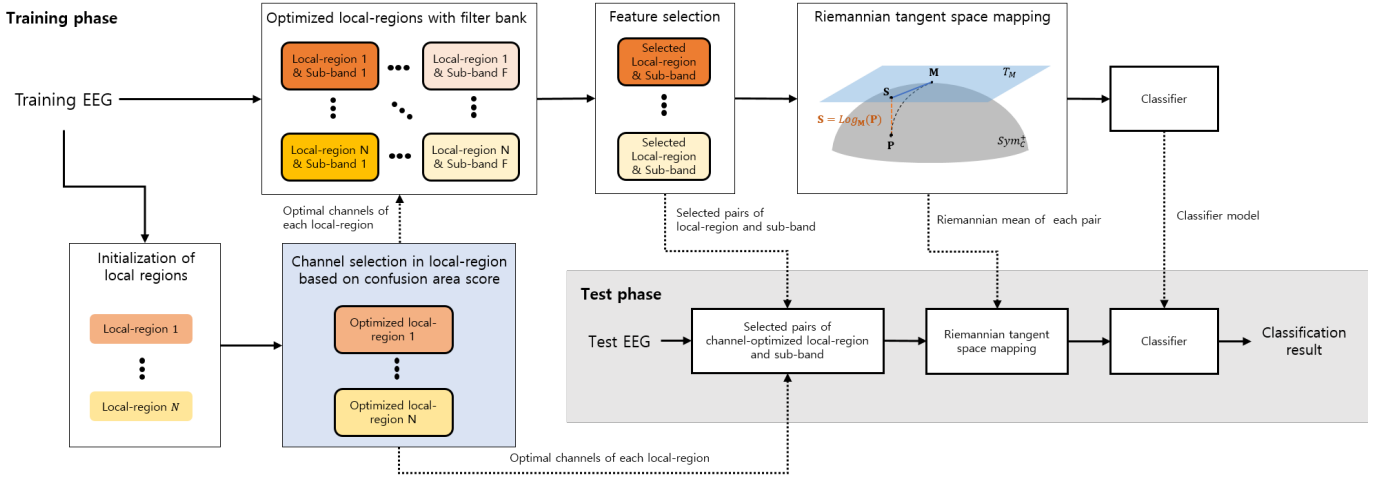


Fig. 1. Overview of the proposed method

optimized local region and filter bank, multiple covariance matrices are computed, and several features are selected based on the confusion area score. The proposed method projects the selected covariance matrices into the Riemannian tangent space, and then applies normalization and vectorization to generate a feature vector. All feature vectors are concatenated and fed to a classifier for classification.

We evaluated the classification accuracy of the proposed method using publicly available BCI Competition III dataset IVa, and the result shows that the proposed method achieves classification accuracy compared to existing related methods.

The paper is organized as follows. In section II, Riemannian geometry, confusion area score, and the proposed method are introduced. Section III provides the data and experiment. Section IV analyzes the results. Finally, the conclusion of this paper is given in Section V.

II. METHOD

A. Riemannian manifold of the covariance matrix

The set of a covariance matrix, which is a symmetric positive definite matrix, lies in a smooth curved space $sym_C^+ = \{P | P \in sym_C, x^T P x < 0 \quad \forall x \in \mathbb{R}^C - \{0\}^C\}$ where $sym_C = \{S | S \in \mathbb{R}^{C \times C}, S = S^T\}$, called a Riemannian manifold. The Riemannian manifold can be locally approximated by tangent space sym_C , which is Euclidean space, defined at a fixed reference point $Q \in sym_C^+$. The logarithmic map $Log_Q : sym_C^+ \rightarrow sym_C$ and exponential map $Exp_Q : sym_C \rightarrow sym_C^+$ enable to project a point between Riemannian manifold and its tangent space, and are defined as

$$Log_Q(P) = S = Q^{\frac{1}{2}} \log_m(Q^{-\frac{1}{2}} P Q^{-\frac{1}{2}}) Q^{\frac{1}{2}} \quad (1)$$

$$Exp_Q(S) = P = Q^{\frac{1}{2}} \exp_m(Q^{-\frac{1}{2}} S Q^{-\frac{1}{2}}) Q^{\frac{1}{2}}, \quad (2)$$

where $\log_m$ and $\exp_m$ are the matrix logarithm and exponential operators [11].

Riemannian distance between $P_i, P_j \in sym_C^+$ is defined as a length of the shortest curve on sym_C^+ and given by the following equation:

$$\begin{aligned} \delta_R^2(P_i, P_j) &= \langle Log_{P_i}(P_j), Log_{P_i}(P_j) \rangle_{P_i} \\ &= trace(\log_m^2(P_i^{-\frac{1}{2}} P_j P_i^{-\frac{1}{2}})). \end{aligned} \quad (3)$$

Riemannian mean M of covariance matrices, $\{P_i | P_i \in sym_C^+, i = 1, \dots, N\}$ where N is the number of covariance matrices, is given by the following equation:

$$M = \arg \min_{P \in sym_C^+} \sum_i \delta_R^2(P, P_i). \quad (4)$$

Since it does not have a closed form solution when $N > 2$, the iterative algorithm is used in general to compute the Riemannian mean [12].

B. Confusion area score: An Riemannian feature evaluation criteria based on Riemannian distance distribution

Confusion area score [10] is designed to evaluate discriminability of Riemannian features based on distributions of Riemannian distance.

For two classes, let $\{P_i^{(1)}\}_{i=1}^{N_1}$ and $\{P_j^{(2)}\}_{j=1}^{N_2}$ be covariance matrices belonging to the first and second class, respectively. The class-dependent Riemannian means are denoted as M_1 and M_2 for each class.

Based on the Riemannian mean $M_a, a \in \{1, 2\}$, the confusion area score considers the distribution of Riemannian distances to $P^{(a')}, a' = a$, and the distribution of Riemannian distance to $P^{(a')}, a' \neq a$.

To assess the degree of overlap between the two distance distributions, two statistical values are utilized: i) the averaged distance from M_a to the $P^{(a')}, a' = a$, which is given by Eq. 6; and ii) the minimum distance from M_a to the $P^{(a')}, a' \neq a$, which is given by Eq. 7.

TABLE I
CHANNELS OF THE OPTIMIZED LOCAL REGION ON BCI COMPETITION III DATASET IV

Number	initial local region	al	aa	av	aw	ay
1	T7,CFC5,C3,CCP5 (4)	T7,CFC5,C3,CCP5 (4)	T7,CFC5,C3,CCP5 (4)	T7,CFC5,C3,CCP5 (4)	CFC5,C3,CCP5 (3)	T7,CFC5,CCP5 (3)
2	CFC5,T7,C3,CFC3 (4)	T7,CFC5,CFC3,C3 (4)	CFC5,CFC3,C3 (3)	T7,CFC5,CFC3,C3 (4)	CFC5,CFC3,C3 (3)	CFC5,CFC3,C3 (3)
3	CFC3,CFC5,C3,C1,Cz (5)	CFC5,CFC3,C3,C1 (4)	CFC5,CFC3,C3,Cz (4)	CFC5,CFC3,C3,C1,Cz (5)	CFC5,CFC3,C3,C1,Cz (5)	CFC5,C3,C1 (3)
4	C3,T7,CFC5,CFC3,CCP5,CCP3,C1 (7)	CFC5,C3,CCP5,CCP3 (4)	CFC5,CFC3,C3,CCP5,CCP3 (5)	CFC3,C3,C1,CCP5,CCP3 (5)	C3,C1,CCP5,CCP3 (4)	CFC5,CFC3,C3,CCP5,CCP3 (5)
5	C1,CFC3,C3,CCP3,Cz (5)	CFC3,C3,C1,CCP3,Cz (5)	CFC3,C3,C1,CCP3 (4)	CFC3,C1,Cz (3)	C3,C1,CCP3,Cz (4)	CFC3,C1,CCP3,Cz (4)
6	CCP5,T7,C3,CCP3,P3 (5)	T7,C3,CCP5,P3 (4)	T7,C3,CCP5,CCP3 (4)	T7,C3,CCP5,CCP3,P3 (5)	T7,C3,CCP5,CCP3 (4)	T7,C3,CCP5,P3 (4)
7	CCP3,C3,C1,CCP5,P3,Cz,Pz (7)	C3,C1,CCP5,Cz (4)	C3,C1,CCP5,P3 (4)	C3,C1,CCP5,CCP3,P3,Pz (6)	C3,C1,CCP3,P3,Cz,Pz (6)	C3,C1,CCP5 (3)
8	P3,CCP5,CCP3,Pz (4)	CCP5,CCP3,Pz (3)	CCP5,CCP3,P3 (3)	CCP5,CCP3,P3 (3)	CCP5,CCP3,P3 (3)	CCP5,P3,Pz (3)
9	Cz,C1,Pz,C2 (4)	C1,Cz,Pz (3)	C1,Cz,C2 (3)	C1,Cz,Pz,C2 (4)	C1,Cz,C2 (3)	C1,Cz,Pz,C2 (4)
10	Pz,CCP3,P3,Cz,CCP4,P4 (6)	CCP3,Cz,Pz (3)	CCP3,P3,Cz,Pz,CCP4,P4 (6)	CCP3,P3,Cz,Pz,P4 (5)	CCP3,P3,Cz,CCP4,P4 (5)	CCP3,Cz,Pz (3)
11	C2,Cz,CFC4,C4,CCP4 (5)	Cz,C2,CFC4,C4 (4)	Cz,C2,CFC4,C4,CCP4 (5)	C2,CFC4,C4,CCP4 (4)	C2,CFC4,C4,CCP4 (4)	Cz,C2,CCP4 (4)
12	CFC4,Cz,C2,CFC6,C4 (5)	Cz,C2,CFC4,CFC6,C4 (5)	C2,CFC4,CFC6,C4 (4)	C2,CFC4,C4 (3)	Cz,C2,CFC4,CFC6,C4 (5)	Cz,C2,CFC4,CFC6 (4)
13	CFC6,CFC4,C4,T8 (4)	CFC4,C4 (2)	CFC4,CFC6,C4 (3)	C4 (1)	CFC4,CFC6,C4 (3)	CFC4,CFC6,C4 (3)
14	C4,C2,CFC4,CFC6,T8,CCP4,CCP6 (7)	C2,CFC4,CFC6,C4 (4)	C2,CFC4,CFC6,C4,T8,CCP4 (6)	C2,CFC4,C4,CCP4,CCP6 (5)	C2,CFC4,C4,CCP4 (4)	C2,CFC4,CFC6,C4,T8,CCP4,CCP6 (7)
15	T8,CFC6,C4,CCP6 (4)	CFC6,C4,T8,CCP6 (4)	CFC6,C4,T8,CCP6 (4)	CCP6 (1)	CFC6,C4,CCP6 (3)	CFC6,C4,T8,CCP6 (4)
16	CCP4,Cz,C2,C4,CCP6,P4,Pz (7)	Cz,C2,C4,CCP6 (4)	Cz,C4,CCP4,CCP6,P4 (5)	Pz,C2,C4,CCP6 (4)	Cz,Pz,C4,CCP4,CCP6,P4 (6)	Cz,C2,C4,CCP4,CCP6 (5)
17	CCP6,C4,T8,CCP4,P4 (5)	C4,T8,CCP6,P4 (4)	C4,CCP4,CCP6,P4 (4)	C4,CCP4,CCP6,P4 (4)	C4,T8,CCP4,CCP6 (4)	C4,T8,CCP6 (3)
18	P4,Pz,CCP4,CCP6 (4)	CCP4,CCP6 (2)	Pz,CCP4,P4 (3)	Pz,CCP4,CCP6 (3)	Pz,CCP4,CCP6,P4 (4)	Pz,CCP4,P4 (3)

TABLE II
EFFECT OF THE CHANNEL-OPTIMIZED LOCAL REGION ON MI CLASSIFICATION ACCURACY(%)

Subject	Local region with filter bank	Channel-optimized local region with filter bank (Proposed)
al	100	100
aa	82.14	83.04
av	71.94	72.45
aw	84.82	87.05
ay	63.10	66.27
mean	80.40	81.76

The normalized difference g^{M_a} of two values is considered as the overlap between class-dependent distributions of Riemannian distance. For example, g^{M_1} is calculated as follows.

$$g^{M_1} = \frac{v_1^{M_1} - p_2^{M_1}}{\delta_R(\mathbf{M}_1, \mathbf{M}_2)} \quad (5)$$

$$\text{where } v_1^{M_1} = \frac{1}{|L_1|} \sum_{i \in \{1, \dots, N_1\}} \delta_R(\mathbf{M}_1, \mathbf{P}_i^{(1)}) \quad (6)$$

$$p_2^{M_1} = \min_{j \in \{1, \dots, N_2\}} \delta_R(\mathbf{M}_1, \mathbf{P}_j^{(2)}) \quad (7)$$

Also, g^{M_2} can be computed in the same manner.

Between two class-dependent scores, the smaller score is selected as the confusion area score g of the Riemannian feature:

$$g = \min\{g^{M_1}, g^{M_2}\} \quad (8)$$

The lower the confusion area score $g \in \mathbb{R}$, the better the discriminability of the covariance matrices.

C. Motor imagery classification using channel-optimized local region Riemannian features with filter bank

An overview of the proposed method is illustrated in Fig 1. The proposed method consists of a local region channel optimization, covariance matrix computation with channel-optimized local region and filter bank, feature selection, and a classification based on tangent space. Details of these components are presented below.

Let $\mathbf{X}_i \in \mathbb{R}^{C \times T}$, $i \in \{1, \dots, N\}$ denote the i -th EEG signal and $y_i \in \{1, 2\}$ denote the corresponding MI class label, where C , T , N , and K are the number of channel, number of EEG time sample per channel, and number of trial, respectively.

For each local region, we select optimal channel subset using the backward iterative algorithm based on the confusion area score. The channel selection process for local region is described in Algorithm 1.

Algorithm 1 Channel selection within local region

Input local region $\mathcal{L}$, labeled covariance matrices $\{P_i, y_i\}_{i=1}^N$

Output channel-optimized local region $\hat{\mathcal{L}}$

$g_0 = \inf$

$g_1 = \text{confusion area score from } \{P_i, y_i\}_{i=1}^N$

$\hat{\mathcal{L}} = \mathcal{L}$

while $|\hat{\mathcal{L}}| > 1$ **do**

for $c \in \hat{\mathcal{L}}$ **do**

$\hat{P}_i = R(P_i, \hat{\mathcal{L}}, c)$: Select rows and columns that corresponds to channels in $\hat{\mathcal{L}} - \{c\}$ from P_i .

$g^c = \text{confusion area score from } \{\hat{P}_i, y_i\}_{i=1}^N$

end for

$g_1 = \min_c g^c$

if $g_0 < g_1$ **then**

 Stop iteration

else

$\hat{\mathcal{L}} = \hat{\mathcal{L}} - \{\arg \min_c g^c\}$

$g_0 = g_1$

end if

end while

The EEG signal $\mathbf{X}_i$ is decomposed into B frequency bands. Let $\mathbf{X}_{i,l,b} \in \mathbb{R}^{C_l \times T}$, $l \in \{1, \dots, C_l\}$, $b \in \{1, \dots, B\}$ denote the i -th trial for the l -th local region and the b -th sub-band, where C_l is the number of selected channel of the l -th local region. The covariance matrix of $\mathbf{X}_{i,l,b}$ is given by

$$\mathbf{P}_{i,l,b} = \frac{1}{T} \mathbf{X}_{i,l,b} \mathbf{X}_{i,l,b}^T \in \text{sym}_+^{C_l} \quad (9)$$

To improve classification performance, several Riemannian features can be selected based on the confusion area score [10]. Let $\mathcal{P}$ be a set of the selected pair of local region and sub-band, and let Q be the number of the selected pair $(l, b) \in \mathcal{P}$.

The proposed method maps $\mathbf{P}_{i,l,b}$ to the tangent space at the Riemannian mean $\mathbf{M}_{l,b}$ and normalized it, as follows.

$$\mathbf{Z}_{i,l,b} = \mathbf{M}_{l,b}^{-\frac{1}{2}} \text{Log}_{\mathbf{M}_{l,b}}(\mathbf{P}_{i,l,b}) \mathbf{M}_{l,b}^{-\frac{1}{2}} \quad (10)$$

TABLE III
COMPARISON OF CLASSIFICATION ACCURACY ON BCI COMPETITION III DATASET IVA

Subject	FBCSP [6]	LRCSP [6]	LRFCSP [7]	TSLDA	LRDM-CAS	Proposed Method
al	98.21	100	100	98.21	100	100 (16)
aa	80.36	74.11	89.29	83.04	78.57	90.18 (4)
av	61.73	67.85	73.47	70.41	76.02	77.04 (3)
aw	88.84	90.07	89.73	94.20	95.09	92.86 (54)
ay	73.81	89.29	94.05	88.10	90.08	91.27 (6)
mean	80.59	84.46	89.31	86.79	87.95	90.27

A local sub-band feature vector $\mathbf{z}_{i,l,b}$ is generated by the matrix vectorization operator $\text{vec} : \text{sym}_C \rightarrow \mathbb{R}^{C(C+1)/2}$.

$$\mathbf{z}_{i,l,b} = \text{vec}(\mathbf{Z}_{i,l,b}), \quad (11)$$

where the matrix vectorization operator $\text{vec}(\cdot)$ is defined as $\text{vec}(\mathbf{A}) := [a_{11}, \sqrt{2}a_{12}, \sqrt{2}a_{13}, \dots, a_{22}, \sqrt{2}a_{23}, \dots, a_{CC}]^\top$.

Finally, the selected Riemannian feature vectors $\{\mathbf{z}_{i,l,b}\}_{(l,b) \in \mathcal{P}}$ are concatenated, and then passed into support vector machine (SVM) for classification.

III. DATA AND EXPERIMENTS

The performance of our proposed method is evaluated using the BCI Competition III dataset IVa [13]. The dataset consists of motor imagery EEG signals for two classes (right hand and foot) recorded from five healthy subjects using 118 channels at a sampling rate of 100 Hz. There are 280 trials per subject and 140 trials per class. The five subjects are referred to as ‘al’, ‘aa’, ‘av’, ‘aw’, ‘ay’ and the number of training trials N for each subjects is 224, 168, 84, 56, and 28, respectively. Detailed information about the dataset can be found in [14].

For the experiments, we use the EEG signal derived from 0.5 to 3 seconds after the visual cue. The signal is band-pass filtered by the 4th order Butterworth filter at 4-40Hz. For feature extraction, 9 filters are used to decompose the frequency range from 4 to 40 Hz with 4 Hz. The proposed method uses $C = 18$ channels as done in [6].

IV. RESULTS AND DISCUSSION

We examine the effect of channel optimization within each local region on each subject. The selected channels in each local region are listed in Table I. The number in parenthesis is the number of channels selected. As can be seen in Table I, the number of selected channels varies from subject to subject, and the selected channels vary across subjects.

To investigate the effect of the channel-optimized local region, we conducted two experiments: one using features from the standard local regions and filter band, and the other using features from the channel-optimized local regions and filter bank. To focus on the effect of channel optimization, all Riemannian features are concatenated without feature selection.

The results are shown in Table II and confirm that the channel-optimized local region achieves performance improvement in all subjects.

The performance of the proposed method is compared to FBCSP [5] LRCSP [6], LRFCSP [7], TSLDA [8], and LRDM-CAS [10]. Table III shows the classification results for each

subject in terms of accuracy rate. The classification results of FBCSP and LRCSP are reported in [6] and the results of LRFCSP is reported in [7]. The classification results of the proposed method are obtained by selecting several optimal features for each subject. The number in parenthesis is the number of selected features.

As shown in Table III, the mean classification accuracy improvements are 9.68%, 5.81%, 0.96%, 3.48% and 2.32% compared to FBCSP, LRCSP, LRFCSP, TSLDA, and LRDM-CAS, respectively. In subjects al, aa and av, the proposed method gives the highest classification accuracy.

V. CONCLUSION

We proposed a novel MI classification method based on local region and Riemannian approach to improve the MI classification performance. The proposed method generates multiple local regions and selects several optimal channels for each local region based on the confusion area score. Using the channel-optimized local regions and filter bank, covariance matrices are computed. Several optimal covariance matrices are selected and projected into the tangent space to be fed into the SVM. The experimental results on the BCI Competition III dataset IVa confirm that the channel-optimized local regions improve the MI classification accuracy and the proposed method achieves performance improvement compared to the existing methods.

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